सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP

Qualification Pack

Electrical Installations

QP Code: ELE/Q6309

Version: 2.0

NSQF Level: 4.5

Electronics Sector Skills Council of India || 155, 2nd Floor, ESC House Okhla Industrial Area-Phase 3
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Contents

ELE/Q6309: Electrical Installations

Brief Job Description

An Electrician, a professional who works on a variety of projects across commercial, residential, agricultural, and industrial sectors. They are responsible for the installation, maintenance, and repair of electrical systems, ensuring high standards of safety and quality. This involves tasks such as planning, designing, selecting and installing electrical systems, testing, fault-finding, and maintaining equipment. Electricians also diagnose malfunctions, program, and commission home and building automation systems.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ELE/N6312: Work organization and management (Electrical Installations)
2. ELE/N6313: Communication and interpersonal skills (Electrical Installations)
3. ELE/N6314: Problem solving, innovation, and creativity (Electrical Installations)
4. ELE/N6315: Planning and design (Electrical Installations)
5. ELE/N6316: Installation (Electrical Installations)
6. ELE/N6317: Testing, reporting, and commissioning (Electrical Installations)
7. ELE/N6318: Maintenance, fault finding, and repair

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	Integration-I&A
Country	India

NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/8212.2401
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	2.0
Reference code on NQR	QG-4.5-EH-05201-2026-V1-ESSCI
NQR Version	2

ELE/N6312: Work organization and management (Electrical Installations)

Description

This NOS unit is about proper training, the correct use of equipment, and adherence to safety protocols in manufacturing settings.

Scope

The scope covers the following :

- Principles and Applications of Safe Working
- Purposes, Uses, Care, and Maintenance of Equipment and Materials
- Environmental and Safety Principles in Good Housekeeping
- Principles of Team Working and Their Applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Follow all health, safety, and environmental rules, procedures, and documentation
PC2. Apply electrical safety principles while carrying out all tasks
PC3. Identify and wear appropriate PPE (safety shoes, gloves, ear and eye protection)
PC4. Select, use, clean, and store tools and equipment correctly and safely
PC5. Select, handle, and store materials safely as per organizational norms
PC6. Protect and handle expensive fixtures and fittings with care
PC7. Maintain a clean, organized, and efficient work area
PC8. Use workflow principles, plan tasks, and prepare the workspace for maximum efficiency
PC9. Measure components, materials, and dimensions accurately as per job requirements
PC10. Manage time effectively to meet turnaround and productivity expectations
PC11. Work efficiently, check progress, and verify quality at every stage
PC12. Apply sustainable practices, minimize wastage, and use ‘green’ materials responsibly
PC13. Adapt to new technologies and integrate improved working methods
PC14. Maintain high standards of quality and accuracy in all working processes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Knowledge of health, safety, and environmental regulations applicable to the workplace
KU2. Understanding of electrical safety principles and hazard prevention methods.
KU3. Understanding of safe tool, equipment, and material handling procedures.
KU4. Understanding of measurement techniques and quality control standards.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to follow safety procedures and comply with organizational guidelines.
GS2. Skill in applying electrical safety measures during task execution.
GS3. Ability to select and properly use PPE as per job requirements.
GS4. Skill in handling, maintaining, and storing tools and materials safely.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Follow all health, safety, and environmental rules, procedures, and documentation	-	-	-	-
PC2. Apply electrical safety principles while carrying out all tasks	-	-	-	-
PC3. Identify and wear appropriate PPE (safety shoes, gloves, ear and eye protection)	-	-	-	-
PC4. Select, use, clean, and store tools and equipment correctly and safely	-	-	-	-
PC5. Select, handle, and store materials safely as per organizational norms	-	-	-	-
PC6. Protect and handle expensive fixtures and fittings with care	-	-	-	-
PC7. Maintain a clean, organized, and efficient work area	-	-	-	-
PC8. Use workflow principles, plan tasks, and prepare the workspace for maximum efficiency	-	-	-	-
PC9. Measure components, materials, and dimensions accurately as per job requirements	-	-	-	-
PC10. Manage time effectively to meet turnaround and productivity expectations	-	-	-	-
PC11. Work efficiently, check progress, and verify quality at every stage	-	-	-	-
PC12. Apply sustainable practices, minimize wastage, and use ‘green’ materials responsibly	-	-	-	-
PC13. Adapt to new technologies and integrate improved working methods	-	-	-	-
PC14. Maintain high standards of quality and accuracy in all working processes	-	-	-	-
NOS Total	-	5	-	-

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6312
NOS Name	Work organization and management (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6313: Communication and interpersonal skills (Electrical Installations)

Description

This NOS unit is about the communication and interpersonal skills.

Scope

The scope covers the following :

- Range and Purposes of Documentation and Publications in Electronic Forms
- Technical Language Associated with the Skill and Technology
- Standards for Routine and Exception Reporting in Oral and Electronic Form
- Required Standards for Communicating with Clients, Team Members, and Others

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Establish and maintain customer confidence through professional behavior and clear communication

PC2. Interpret customer requirements accurately and manage expectations effectively

PC3. Provide guidance and recommendations on suitable products, solutions, and technological advancements

PC4. Visualize customer needs and propose design improvements within their budget

PC5. Use effective questioning techniques to understand customer requirements thoroughly

PC6. Provide clear, accurate, and actionable instructions to customers and colleagues

PC7. Coordinate with related trades and introduce them when needed to meet customer requirements

PC8. Prepare accurate written reports for customers and organizational documentation

PC9. Produce realistic cost estimates and time schedules for customer tasks

PC10. Adapt to the changing needs and workflows of related trades

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the importance of professional conduct and effective communication in building long-term customer trust and confidence.

KU2. Possess knowledge of methods to accurately interpret customer requirements and manage expectations realistically.

KU3. Understand product specifications, technological advancements, and solution-based approaches to provide appropriate recommendations.

KU4. Have knowledge of basic design principles, budgeting considerations, and cost estimation techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate clear and confident communication skills while interacting with customers and colleagues.
GS2. Apply analytical and questioning skills to accurately identify and visualize customer needs.
GS3. Prepare precise cost estimates, time schedules, and professional written reports.
GS4. Coordinate effectively with related trades to ensure seamless task execution.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**5**	**-**	**-**
PC1. Establish and maintain customer confidence through professional behavior and clear communication	-	-	-	-
PC2. Interpret customer requirements accurately and manage expectations effectively	-	-	-	-
PC3. Provide guidance and recommendations on suitable products, solutions, and technological advancements	-	-	-	-
PC4. Visualize customer needs and propose design improvements within their budget	-	-	-	-
PC5. Use effective questioning techniques to understand customer requirements thoroughly	-	-	-	-
PC6. Provide clear, accurate, and actionable instructions to customers and colleagues	-	-	-	-
PC7. Coordinate with related trades and introduce them when needed to meet customer requirements	-	-	-	-
PC8. Prepare accurate written reports for customers and organizational documentation	-	-	-	-
PC9. Produce realistic cost estimates and time schedules for customer tasks	-	-	-	-
PC10. Adapt to the changing needs and workflows of related trades	-	-	-	-
NOS Total	**-**	**5**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6313
NOS Name	Communication and interpersonal skills (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6314: Problem solving, innovation, and creativity (Electrical Installations)

Description

This NOS unit is about identifying problems and addressing them promptly.

Scope

The scope covers the following :

- Identify Cross-Trade Issues
- Swift Problem Recognition & Resolution
- Contribute Ideas for Improvement
- Alternative Solutions for Customers

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Check work regularly to identify errors early and prevent issues at later stages

PC2. Identify problems originating from related trades (e.g., heating system, ventilation, pump installation)

PC3. Challenge unclear or incorrect information to avoid future complications

PC4. Recognize and analyze problems using systematic diagnostic approaches

PC5. Apply a self-managed, logical process to resolve issues efficiently

PC6. Stay updated with industry trends, new technologies, standards, and smart home solutions

PC7. Identify procurement-related issues and recommend alternative materials or methods

PC8. Suggest innovative and practical solutions to improve customer satisfaction

PC9. Demonstrate willingness to adopt new working methods, tools, and ready-made components

PC10. Recommend smarter, more cost-efficient, and sustainable installation options to customers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand systematic inspection methods to detect errors early and prevent future complications.

KU2. Possess knowledge of interdependencies between related trades such as heating, ventilation, and pump installations.

KU3. Understand diagnostic principles and logical problem-solving approaches for effective troubleshooting.

KU4. Stay informed about industry standards, emerging technologies, smart home solutions, and sustainable practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate strong analytical skills to identify, assess, and resolve technical issues independently.

GS2. Communicate assertively to clarify unclear or incorrect information and prevent project risks.

GS3. Apply innovative thinking to recommend practical and customer-focused solutions.

GS4. Adapt proactively to new tools, working methods, and technological advancements.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Check work regularly to identify errors early and prevent issues at later stages	-	-	-	-
PC2. Identify problems originating from related trades (e.g., heating system, ventilation, pump installation)	-	-	-	-
PC3. Challenge unclear or incorrect information to avoid future complications	-	-	-	-
PC4. Recognize and analyze problems using systematic diagnostic approaches	-	-	-	-
PC5. Apply a self-managed, logical process to resolve issues efficiently	-	-	-	-
PC6. Stay updated with industry trends, new technologies, standards, and smart home solutions	-	-	-	-
PC7. Identify procurement-related issues and recommend alternative materials or methods	-	-	-	-
PC8. Suggest innovative and practical solutions to improve customer satisfaction	-	-	-	-
PC9. Demonstrate willingness to adopt new working methods, tools, and ready-made components	-	-	-	-
PC10. Recommend smarter, more cost-efficient, and sustainable installation options to customers	-	-	-	-
NOS Total	-	5	-	-

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6314
NOS Name	Problem solving, innovation, and creativity (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6315: Planning and design (Electrical Installations)

Description

This NOS unit is about read, interpret, and revise technical drawings and documentation, including layout and circuit diagrams, to ensure accuracy.

Scope

The scope covers the following :

- Read and Interpret Drawings
- Revise Drawings and Documentation
- Follow Written Instructions
- Plan Installation Work

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand and apply different types of standards, codes, and regulations relevant to installations

PC2. Read, interpret, and revise technical drawings, layout plans, and circuit diagrams accurately

PC3. Follow written instructions and manuals to ensure correct installation procedures

PC4. Identify suitable materials and installation techniques for different environments and conditions

PC5. Plan installation work efficiently using drawings, documentation, and resources provided

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand various standards, codes, and regulatory requirements applicable to installation work.

KU2. Possess knowledge of technical drawings, layout plans, and circuit diagrams for accurate interpretation.

KU3. Understand installation manuals and written instructions to ensure compliance with prescribed procedures.

KU4. Have knowledge of suitable materials and installation techniques for different environmental conditions.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Apply standards and regulations accurately during installation activities.

GS2. Demonstrate proficiency in reading, interpreting, and modifying technical drawings and diagrams.

GS3. Follow written guidelines and manuals carefully to ensure quality and safety.

GS4. Select appropriate materials and techniques based on site conditions and project requirements.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**10**	**-**	**-**
PC1. Understand and apply different types of standards, codes, and regulations relevant to installations	-	-	-	-
PC2. Read, interpret, and revise technical drawings, layout plans, and circuit diagrams accurately	-	-	-	-
PC3. Follow written instructions and manuals to ensure correct installation procedures	-	-	-	-
PC4. Identify suitable materials and installation techniques for different environments and conditions	-	-	-	-
PC5. Plan installation work efficiently using drawings, documentation, and resources provided	-	-	-	-
NOS Total	**-**	**10**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6315
NOS Name	Planning and design (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6316: Installation (Electrical Installations)

Description

This NOS unit is about the selection and installation of equipments, wireways, and ducting systems according to drawings and industrial standards.

Scope

The scope covers the following :

- Select and Install Equipment & Wire Ways
- Install Ducting & Cabling Systems
- Select and Install Cables
- Secure Cable Installation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Select suitable ducting and wiring systems for commercial, domestic, residential, agricultural, and industrial applications

PC2. Identify and choose appropriate electrical switchboards based on environment and load requirements

PC3. Select suitable lighting and heating systems for different sectors

PC4. Choose correct control devices, socket outlets, and smart building components based on use-case

PC5. Install equipment and wireways according to drawings and job documentation

PC6. Install ducting and cabling systems on various surfaces as per manufacturer instructions and standards

PC7. Select and install single/double insulated cables inside ducts, conduits, and flexible conduits

PC8. Fix double insulated cables securely onto cable trays, ladders, and surfaces as per standards

PC9. Measure, cut, and assemble metal/plastic ducting (trunking) accurately without distortion

PC10. Assemble termination adaptors, including glands, and secure ducts onto surfaces

PC11. Install metal/plastic conduits and flexible conduits with correct bends and secure mounting

PC12. Use proper termination adaptors for conduit entry into boards, boxes, and ducts

PC13. Install cable ladders and trays securely on surfaces following industrial standards

PC14. Install and assemble electrical switchboards, including switches, RCDs, MCBs, fuses, relays, timers, and automation devices

PC15. Connect wiring, structured cabling systems, EV charging systems, solar panels, and other sustainable systems according to standards and manufacturer instructions

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the selection criteria for ducting, wiring systems, switchboards, lighting, heating, and control devices across various sectors.

KU2. Possess knowledge of cable types, insulation standards, and installation methods for different environments.

KU3. Understand the principles of assembling and installing conduits, trunking, cable trays, and ladders as per industrial standards.

KU4. Have knowledge of electrical switchboard components such as RCDs, MCBs, fuses, relays, timers, and automation devices.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate the ability to accurately select and install appropriate materials and systems based on application requirements.

GS2. Measure, cut, assemble, and mount ducting, conduits, and cable management systems precisely and securely.

GS3. Install and terminate cables safely and efficiently following technical drawings and job documentation.

GS4. Assemble and wire electrical switchboards and associated components with accuracy and compliance.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**35**	**-**	**-**
PC1. Select suitable ducting and wiring systems for commercial, domestic, residential, agricultural, and industrial applications	-	-	-	-
PC2. Identify and choose appropriate electrical switchboards based on environment and load requirements	-	-	-	-
PC3. Select suitable lighting and heating systems for different sectors	-	-	-	-
PC4. Choose correct control devices, socket outlets, and smart building components based on use-case	-	-	-	-
PC5. Install equipment and wireways according to drawings and job documentation	-	-	-	-
PC6. Install ducting and cabling systems on various surfaces as per manufacturer instructions and standards	-	-	-	-
PC7. Select and install single/double insulated cables inside ducts, conduits, and flexible conduits	-	-	-	-
PC8. Fix double insulated cables securely onto cable trays, ladders, and surfaces as per standards	-	-	-	-
PC9. Measure, cut, and assemble metal/plastic ducting (trunking) accurately without distortion	-	-	-	-
PC10. Assemble termination adaptors, including glands, and secure ducts onto surfaces	-	-	-	-
PC11. Install metal/plastic conduits and flexible conduits with correct bends and secure mounting	-	-	-	-
PC12. Use proper termination adaptors for conduit entry into boards, boxes, and ducts	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Install cable ladders and trays securely on surfaces following industrial standards	-	-	-	-
PC14. Install and assemble electrical switchboards, including switches, RCDs, MCBs, fuses, relays, timers, and automation devices	-	-	-	-
PC15. Connect wiring, structured cabling systems, EV charging systems, solar panels, and other sustainable systems according to standards and manufacturer instructions	-	-	-	-
NOS Total	**-**	**35**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6316
NOS Name	Installation (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	6
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6317: Testing, reporting, and commissioning (Electrical Installations)

Description

This NOS unit is about testing the installations for safety.

Scope

The scope covers the following :

- Pre-Energization Testing
- Post-Energization Testing
- Set-Up of Equipment
- Ensure Full Functionality

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand and apply industrial regulations and standards relevant to different types of electrical installations

PC2. Use appropriate verification standards, methods, and reporting techniques to record results accurately

PC3. Identify and correctly use different types of measuring instruments

PC4. Select and operate tools and software required for parameterization, programming, and commissioning of electrical systems

PC5. Ensure electrical installations operate according to planned specifications and customer requirements

PC6. Conduct pre-energization tests including insulation resistance, earth continuity, correct polarity, and visual inspections for safety

PC7. Conduct post-energization testing to verify proper functioning of all installed equipment, correct voltage, phase rotation, and protection devices

PC8. Program and set up equipment such as programmable relays, bus-systems, timers, and overload relays according to job requirements

PC9. Ensure installations are fully functional and customers are able to operate them safely and efficiently

PC10. Provide accurate data to update drawings, "as-built" documentation, and other related records after installation completion

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand industrial regulations, compliance standards, and verification procedures applicable to electrical installations.

KU2. Possess knowledge of testing methods, reporting techniques, and documentation standards for accurate result recording.

KU3. Understand the functions and correct usage of electrical measuring instruments and diagnostic tools.

KU4. Have knowledge of parameterization, programming, and commissioning processes for programmable relays, bus systems, and automation devices.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate the ability to conduct systematic testing, inspection, and verification of electrical installations.

GS2. Operate measuring instruments, tools, and programming software accurately and safely.

GS3. Perform insulation resistance, continuity, polarity, voltage, and phase rotation tests effectively.

GS4. Program, configure, and commission electrical equipment according to job and customer requirements.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**25**	**-**	**-**
PC1. Understand and apply industrial regulations and standards relevant to different types of electrical installations	-	-	-	-
PC2. Use appropriate verification standards, methods, and reporting techniques to record results accurately	-	-	-	-
PC3. Identify and correctly use different types of measuring instruments	-	-	-	-
PC4. Select and operate tools and software required for parameterization, programming, and commissioning of electrical systems	-	-	-	-
PC5. Ensure electrical installations operate according to planned specifications and customer requirements	-	-	-	-
PC6. Conduct pre-energization tests including insulation resistance, earth continuity, correct polarity, and visual inspections for safety	-	-	-	-
PC7. Conduct post-energization testing to verify proper functioning of all installed equipment, correct voltage, phase rotation, and protection devices	-	-	-	-
PC8. Program and set up equipment such as programmable relays, bus-systems, timers, and overload relays according to job requirements	-	-	-	-
PC9. Ensure installations are fully functional and customers are able to operate them safely and efficiently	-	-	-	-
PC10. Provide accurate data to update drawings, “as-built” documentation, and other related records after installation completion	-	-	-	-
NOS Total	**-**	**25**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6317
NOS Name	Testing, reporting, and commissioning (Electrical Installations)
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

ELE/N6318: Maintenance, fault finding, and repair

Description

This NOS unit is about troubleshooting and diagnosing faults in electrical installations

Scope

The scope covers the following :

- Adapt to Changing Circumstances
- Troubleshoot Electrical Installations
- Diagnose Electrical Installation Issues
- Verify Compliance with Current Standards:

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand different types and generations of electrical installations and their purposes

PC2. Adapt work practices to changing circumstances and customer requirements

PC3. Troubleshoot electrical installations to identify faults such as short/open circuits, incorrect polarity, insulation resistance issues, earth continuity faults, incorrect equipment settings, and programming errors

PC4. Diagnose complex installation problems including bad connections, incorrect wiring, high loop impedance, and equipment failure

PC5. Verify that existing installations comply with current electrical standards and regulations

PC6. Use, test, and calibrate measuring instruments including insulation testers, continuity testers, multimeters, clamp meters, and network cable testers

PC7. Repair and replace faulty components in electrical installations safely and efficiently

PC8. Rewire or repair faulty installations to restore proper functionality

PC9. Recycle and dispose of replaced components in a sustainable and environmentally responsible manner

PC10. Maintain documentation of faults, repairs, and actions taken for customer and organizational records

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand different generations and types of electrical installations and their functional applications.

KU2. Possess knowledge of common electrical faults such as short/open circuits, polarity errors, insulation resistance issues, and earth continuity faults.

KU3. Understand advanced diagnostic principles for identifying complex issues like high loop impedance, bad connections, and equipment failures.

KU4. Have knowledge of current electrical standards and compliance requirements for existing installations.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate the ability to troubleshoot and diagnose electrical faults using systematic testing methods.

GS2. Adapt work practices effectively according to changing site conditions and customer requirements.

GS3. Safely repair, replace, and rewire faulty components to restore system functionality.

GS4. Accurately use and test measuring instruments such as multimeters, clamp meters, and insulation testers.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	**15**	**-**	**-**
PC1. Understand different types and generations of electrical installations and their purposes	-	-	-	-
PC2. Adapt work practices to changing circumstances and customer requirements	-	-	-	-
PC3. Troubleshoot electrical installations to identify faults such as short/open circuits, incorrect polarity, insulation resistance issues, earth continuity faults, incorrect equipment settings, and programming errors	-	-	-	-
PC4. Diagnose complex installation problems including bad connections, incorrect wiring, high loop impedance, and equipment failure	-	-	-	-
PC5. Verify that existing installations comply with current electrical standards and regulations	-	-	-	-
PC6. Use, test, and calibrate measuring instruments including insulation testers, continuity testers, multimeters, clamp meters, and network cable testers	-	-	-	-
PC7. Repair and replace faulty components in electrical installations safely and efficiently	-	-	-	-
PC8. Rewire or repair faulty installations to restore proper functionality	-	-	-	-
PC9. Recycle and dispose of replaced components in a sustainable and environmentally responsible manner	-	-	-	-
PC10. Maintain documentation of faults, repairs, and actions taken for customer and organizational records	-	-	-	-
NOS Total	**-**	**15**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6318
NOS Name	Maintenance, fault finding, and repair
Sector	Electronics
Sub-Sector	
Occupation	Integration-I&A
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).

4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.

5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6312.Work organization and management (Electrical Installations)	-	5	-	-	5	5
ELE/N6313.Communication and interpersonal skills (Electrical Installations)	-	5	-	-	5	5
ELE/N6314.Problem solving, innovation, and creativity (Electrical Installations)	-	5	-	-	5	5
ELE/N6315.Planning and design (Electrical Installations)	-	10	-	-	10	10
ELE/N6316.Installation (Electrical Installations)	-	35	-	-	35	35
ELE/N6317.Testing, reporting, and commissioning (Electrical Installations)	-	25	-	-	25	25
ELE/N6318.Maintenance, fault finding, and repair	-	15	-	-	15	15
Total	**0**	**100**	**-**	**-**	**100**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.